

ADVANCED STATISTICAL & CALIBRATION METRICS

Arabic Fake News Detection – Model Trust Analysis

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— MODULE 01 —

WHY NOT JUST ACCURACY?

The problem with vanity metrics

\$ Accuracy is Not Enough

— The Classic Trap

▶ Dataset: 90% real, 10% fake → dumb model gets 90% by always saying 'real'

— What we actually need to know:

▶ Does the model's confidence actually match reality?

▶ If it says '99% fake' – is it right 99% of the time?

▶ # # This is called Probability Calibration

▶ → The model must be honest, not just accurate

— MODULE 02 —

KEY METRICS

What we measure and why

\$ The 4 Core Metrics

METRIC

- ▶ MCC
- ▶ ECE
- ▶ Brier Score
- ▶ Bootstrap CI

WHAT IT ANSWERS

- ▶ Did the model do well on BOTH classes?
- ▶ Is the confidence gap small or large?
- ▶ Does it pay for being confidently wrong?
- ▶ Are these results stable – or just luck?

RESULTS SUMMARY

Numbers first – then we break them down graph by graph

\$ How Did the Model Do?

Accuracy → 0.926

Balanced Acc → 0.929

F1 Score → 0.920

MCC → 0.853

Brier Score → 0.058 (lower = better)

ECE → 0.045 (lower = better)

All numbers confirmed stable over 1000 bootstrap runs

— GRAPH 01 —

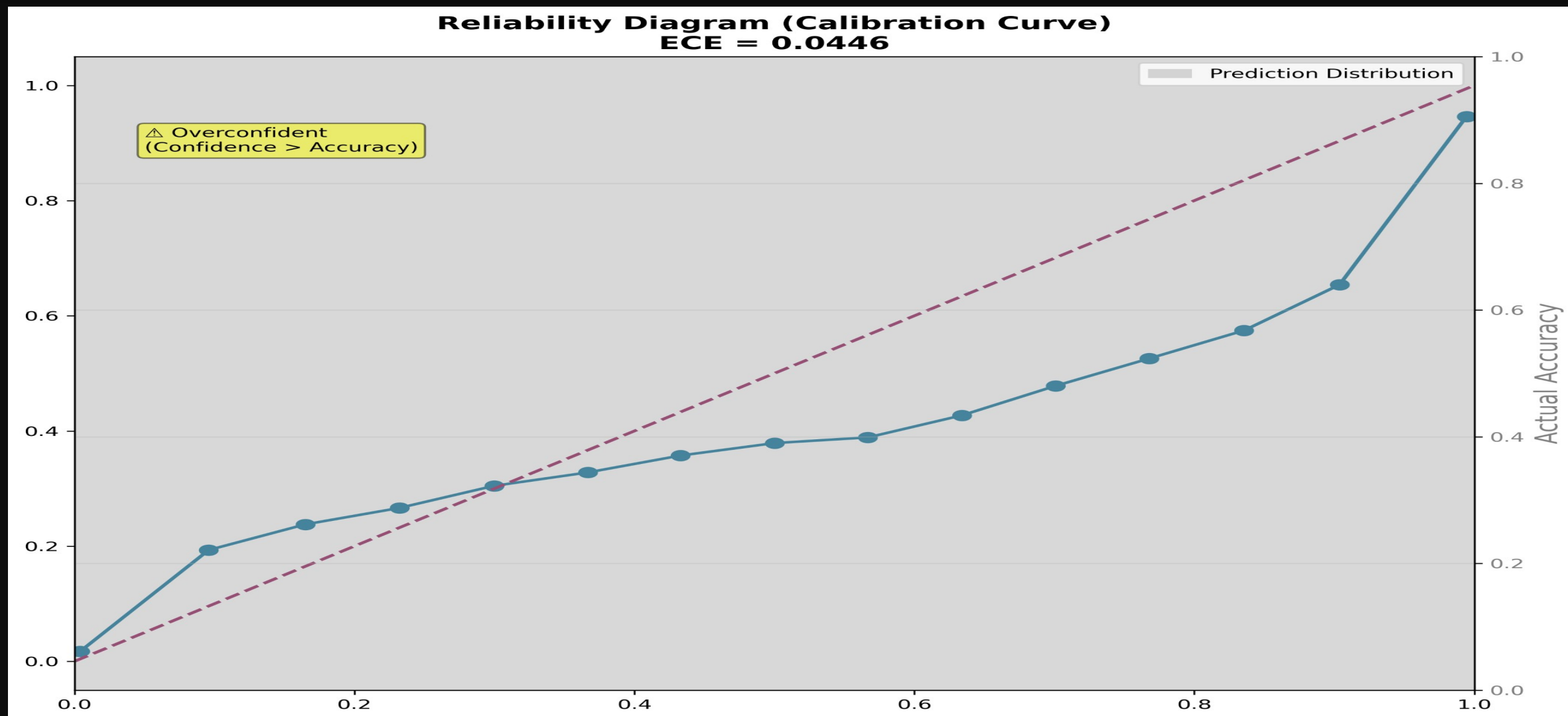
RELIABILITY DIAGRAM

Is the model honest about its confidence?

\$ How to Read the Reliability Diagram

- ▶ X-axis = Predicted Confidence | Y-axis = Actual Accuracy
- ▶ Blue line = our model | Pink dashed = perfect calibration (45°)
- ▶ Line BELOW 45° → Overconfident (claims more certainty than it earns)
- ▶ Line ABOVE 45° → Underconfident (performs better than it claims)
- ▶ ECE = 0.0446 → overall gap is small → well-calibrated overall
- ▶ # # Trust it most at extremes – be cautious in the 0.3–0.85 range

Reliability Diagram (Calibration Curve) – ECE = 0.0446



Blue line below diagonal = overconfident in mid-range | Recovers strongly near confidence 1.0

— GRAPH 02 —

CALIBRATION BY BIN

Per-bin accuracy vs confidence – plus sample distribution

\$ What calibration_by_bin.png Shows

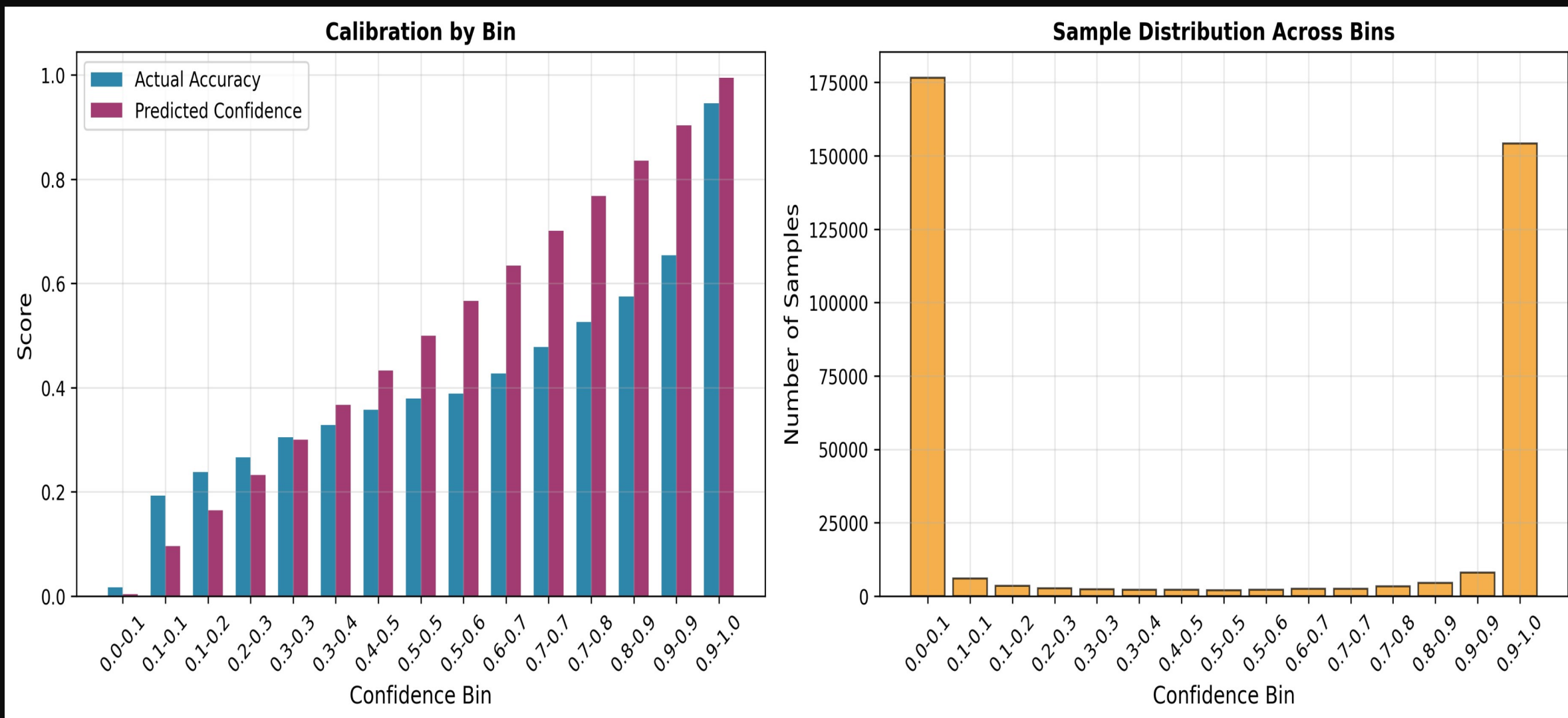
— Left Panel — Calibration by Bin

- ▶ Pink bar (Predicted Confidence) taller than Blue bar (Actual Accuracy) → overconfident
- ▶ Bins 0.9–1.0: Blue (~0.94) nearly matches Pink (~0.98) — earns it here

— Right Panel — Sample Distribution

- ▶ Bin 0.0–0.1: ~175,000 samples | Bin 0.9–1.0: ~155,000 samples
- ▶ Middle bins (0.1–0.9): nearly empty — model is decisively binary
- ▶ # # Mid-range overconfidence barely affects ECE — almost no samples land there

Calibration by Bin + Sample Distribution



Left: Pink vs Blue per bin | Right: Bimodal distribution – almost nothing in the middle

— GRAPH 03 —

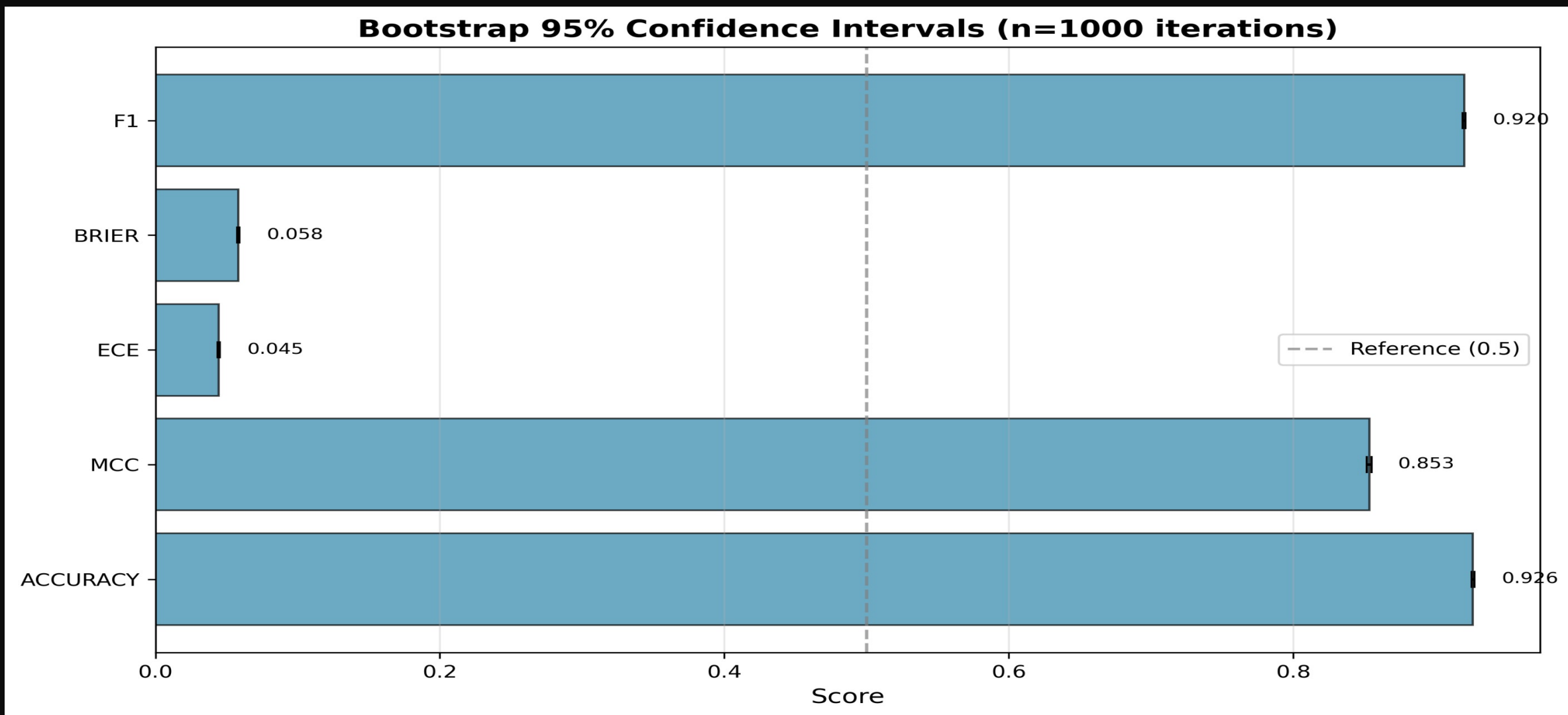
BOOTSTRAP 95% CI

Proving these results are not just luck

\$ What the Bootstrap Graph Shows

- ▶ Evaluation repeated 1000 times on random sub-samples of the data
- ▶ Short error bars = stable result – not sensitive to sample selection
- ▶ ACCURACY = 0.926 → extremely tight error bar
- ▶ MCC = 0.853 → 95% CI: [0.8518, 0.8551] (only 0.003 wide!)
- ▶ ECE = 0.045 → confirmed stable
- ▶ # # This rules out luck – the model is genuinely this good

Bootstrap 95% Confidence Intervals (n=1000 iterations)



Accuracy 0.926 | F1 0.920 | MCC 0.853 | Brier 0.058 | ECE 0.045 – all with tight error bars

— GRAPH 04 —

CONFIDENCE HISTOGRAM

Who is confident – and are they actually right?

\$ What confidence_histogram.png Shows

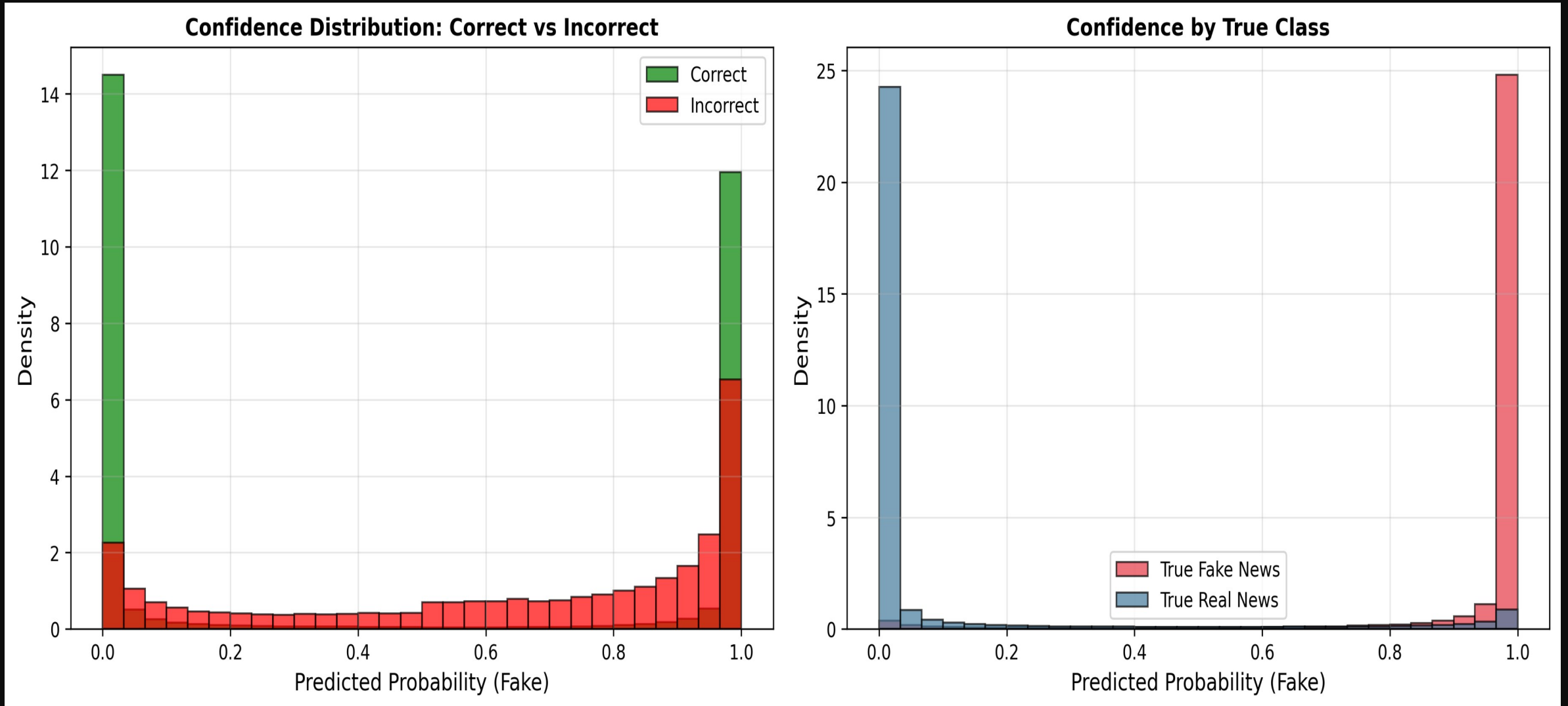
— Left — Correct vs Incorrect

- ▶ Green (correct) spikes at 0.0 and near 1.0 — right when it's sure
- ▶ Red (incorrect) scattered in middle — wrong = uncertain

— Right — Confidence by True Class

- ▶ Blue (True Real News) → peaks near 0.0 — correctly low $P(\text{Fake})$
- ▶ Pink (True Fake News) → peaks near 1.0 — correctly high $P(\text{Fake})$
- ▶ # # The two classes barely overlap — clean decision boundary

Confidence Distribution: Correct vs Incorrect / By True Class



Model commits to extremes – correct predictions cluster at 0 and 1, errors live in the middle

— GRAPH 05 —

CALIBRATION DASHBOARD

Everything in one 9-panel summary

\$ Key Panels to Focus On

- Bubble Chart (middle center)
 - ▶ Giant bubble at top-right: high confidence + high accuracy – where most predictions land
- ECE Contribution per Bin (top middle)
 - ▶ Largest bars at 0.7–0.8 bins, but those bins are nearly empty → ECE stays low
- Summary Boxes (bottom row)
 - ▶ Brier=0.058 | MCC=0.853 | Acc=0.926 | Balanced Acc=0.929 | F1=0.920
 - ▶ # # Balanced accuracy > raw accuracy = no class bias detected

Advanced Calibration & Statistical Metrics Dashboard

9 panels: Reliability Diagram · ECE Contribution · Bootstrap CI · Bubble Chart · Brier · MCC · Performance Summary

— CONCLUSION —

FINAL VERDICT

What does all of this actually mean?

\$ Key Takeaways

- The model works – and we proved it statistically
 - ▶ 92.6% accuracy · 0.853 MCC · stable across 1000 bootstrap runs
- It's honest when it's sure
 - ▶ High-confidence predictions are genuinely reliable – trust them
- Mid-range overconfidence exists but doesn't matter in practice
 - ▶ Almost no predictions land in the 0.1–0.9 range anyway
 - ▶ # # Production tip: confidence < 70% → flag for human reviewer

[EOF]

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